Jayanta Dey

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U.S. Permanent Resident — Outstanding Professor/Researcher (EB-1B)

SUMMARY

Machine Learning Research Scientist with a Johns Hopkins PhD in Biomedical Engineering developing adaptive, memory-based AI and clinically grounded machine learning systems. My fundamental research asks how intelligent systems can continuously compress experience into useful memory, retain knowledge under distribution shift, and achieve long-range context without storing all past data—spanning continual learning, memory consolidation, and non-stationary learning. In parallel, I develop clinical AI methods for biomedical imaging, uncertainty and calibration, and LLM-based clinical decision support, currently collaborating with trauma surgeons on IRB-reviewed multicenter clinical data. Author of 16 publications (250+ citations) in TPAMI, TMLR, CoLLAs, and biomedical venues.

EDUCATION

Johns Hopkins University, Baltimore, MD
PhD, Biomedical Engineering, 2024
MS, Biomedical Engineering (earned en route to PhD)

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh
MS, Electrical Engineering, 2019
BS, Electrical Engineering, 2017

SELECTED PEER-REVIEWED PUBLICATIONS (6 of 16)

1. **Dey, J.**, Srivastava, S., Lerner, I., Kanan, C., & Kudithipudi, D. (2026). *SHARP: Sleep-based Hierarchical Accelerated Replay for Long-Range Non-Stationary Temporal Pattern Recognition*. **Conference on Lifelong Learning Agents (CoLLAs 2026)**. **Selected for oral presentation (Top 4% of Submissions)**.
2. Vogelstein, J. T.*, **Dey, J.***, Helm, H. S., LeVine, W., Mehta, R. D., Tomita, T. M., ... & Priebe, C. E. (2025). Simple lifelong learning machines. **IEEE Transactions on Pattern Analysis and Machine Intelligence**. (**Co-first author with PhD advisor; corresponding author**)
3. **Dey, J.**, Xu, H., De Silva, A., & Vogelstein, J. T. (2025). Simple Calibration via Geodesic Kernels. **Transactions on Machine Learning Research**.
4. Liu, T*, **Dey, J.***, Xu, B*, Bridgeford, EW, Alldritt, S, Nenning, K-H, et al. (2026) Statistically valid explainable black-box machine learning: applications in sex classification across species using brain imaging. **PLoS One** 21(4): e0346575. (**Co-first author**)
5. **Dey, J**., Hossain, M. S. B., & Haque, M. A. (2018, December). An ensemble SVM-based approach for voice activity detection. In 2018 **10th International Conference on Electrical and Computer Engineering (ICECE)** (pp. 297-300).
6. **Dey, J.**, & Hasan, M. K. (2018). Ultrasonic tissue reflectivity function estimation using correlation constrained multichannel FLMS algorithm with missing rf data. **Biomedical Physics & Engineering Express**, 4(4), 045024.

SKILLS

Machine Learning: Multimodal Continual Learning, Memory-Augmented AI, Agent Memory, Long-Range Sequence Modeling, Streaming Learning, Transfer Learning, Representation Learning, Uncertainty Quantification, OOD Generalization, Interpretable AI, Neurosymbolic AI

Healthcare & Biomedical AI: Clinical Decision Support, Medical Imaging, Ultrasound RF Reconstruction, Biomedical Signal Processing, LLM-based Clinical Reasoning, Trauma AI, RAG, Causal Analysis, HIPPA, IRB
Programming: Python, SQL, R, MATLAB, C/C++
Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, pandas, LangChain, OpenAI API, Hugging Face Transformers, Pydantic
Tools & Infrastructure: Git, version control, unit testing, Docker, CUDA, AWS, Azure, CircleCI, Codecov

WORK EXPERIENCE

NuAI Lab, UTSA | San Antonio, TX
Postdoctoral Research Scientist(Computer Engineering) | July 2024 – Present

- **Sleep-Inspired Foundation Model:** Developed SHARP, **a novel sleep-inspired memory augmented architecture** for efficient long-range, non-stationary learning in collaboration with neuroscientists and ML researchers, achieving 12.3× faster runtime than a comparable 5M-parameter Transformer; selected for **CoLLAs 2026 Oral**.
- **Clinical AI & LLMs:** Collaborating with trauma surgeons and healthcare startup **Leah Health** on an LLM-based clinical study using **IRB-reviewed data from a seven-center Texas trauma network** to model clinician disagreement, identify high-complexity cases, and prioritize expert review.
- **Cross-Disciplinary Clinical Research:** Advising a multi-center trauma-AI program involving 6 trauma surgeons and $2.5M in secured funding, including work on clinician digital twins and expert-disagreement biomarkers for clinical complexity.

NeuroData Lab, Johns Hopkins University | Baltimore, MD
Graduate Researcher (Ph.D. Candidate) | July 2019 – June 2024

- **Multimodal Continual Learning:** Led DARPA L2M + Microsoft Research research that introduced forward/backward transfer metrics and a continual learning algorithm, SiLLy, outperforming continual-learning baselines across vision, speech, and tabular tasks.
- **Model Reliability & Validation:** Developed uncertainty-calibration methods and rigorously validated model reliability across **45+ datasets** under distribution shift.
- **AI Reliability & Explainability:** Developed statistically valid methods for interpreting and auditing black-box ML models across biomedical imaging datasets.
- **Open-Source ML:** Developed and maintained open-source ML/statistics packages (Proglearn, hyppo) with unit testing, CI/CD, GPU/cloud computing, and reproducible research workflows.

DSPRL Lab, BUET | Dhaka, Bangladesh
Research Assistant (Electrical Engineering) | January 2016 – July 2019

- **Biomedical Signal Processing:** Developed ultrasound RF reconstruction and adaptive despeckling algorithms using multichannel signal processing to improve diagnostic B-mode image quality.
- **Medical Imaging Simulation:** Built FIELD-II simulations with 3D tissue phantoms and realistic transducer configurations to evaluate imaging algorithms under varying sampling, scatterer density, and noise; resulting in 2 publications.

AWARDS & HONORS

IEEE Signal Processing Cup champion
AWS Public Sector Cloud Credit Grant, *Amazon Web Services* (Supported the proglearn package)
Ranked 4th, 3.95/4.0, *Bangladesh University of Engineering and Technology*
Regional Champion, *Bangladesh Mathematical Olympiad,* 2008, 2010